CardamomOT — analysis report

/Users/eventre/Documents/GitHub/CardamomOT/experimental_datasets/Semrau

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Run

Split	train
Stimulus penalisation	1.0
Prior penalisation	1.0
Genes	41
Stimuli	1
Observed cells	1714
Timepoints	0, 6, 12, 24, 36, 48, 60, 72, 96
Cell types	Pluripotency, Post-implantation_Epiblast, Extraembryonic_Endoderm, Neuroectoderm
Network shown	inter_simul.npy
Embeddings	UMAP; mRNA: log1p=True, normalise=False, cell_depth_for_representation=False (raw counts, model draws at the depth of the cells)

Perturbations (perturbation_simulation)

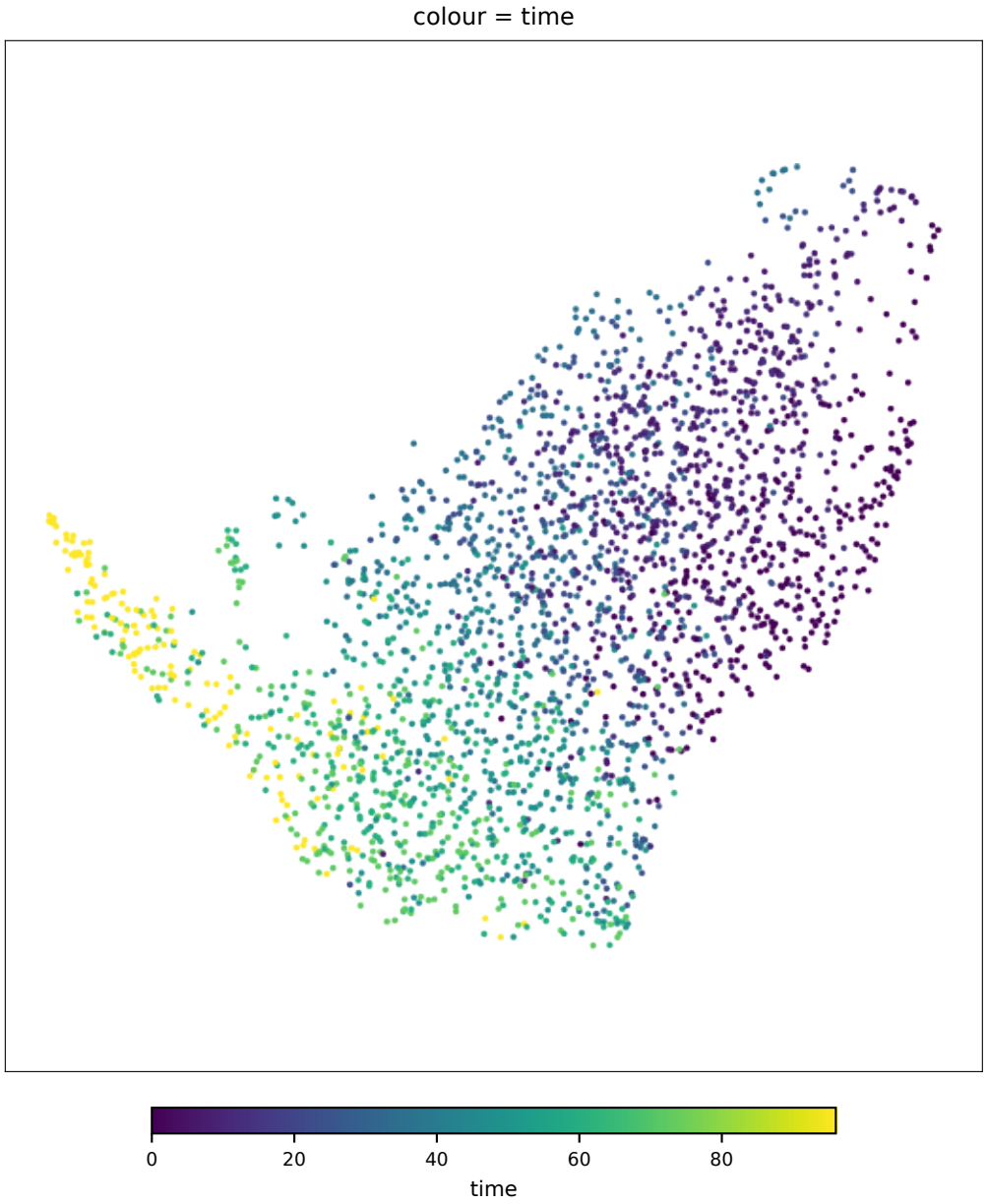
- ✓ P1 KO Dnmt3a
- ✓ P2 OV Dnmt3a

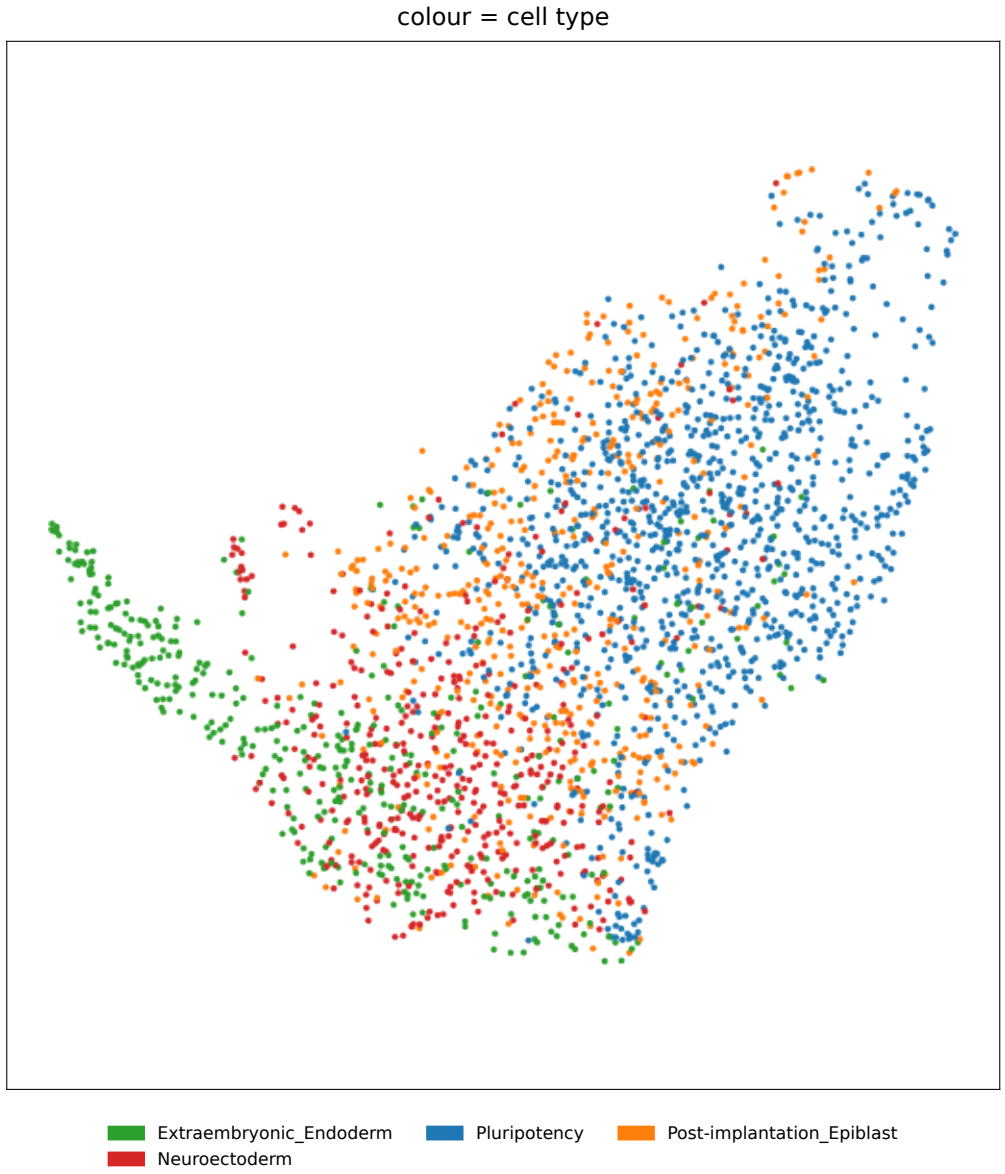
Contents

Data — UMAP of all the cells of Data/data.h5ad (time, samples), also on the genes of the model with select_genes; teaser: classical OT vs CardamomOT displacement fields (
1. Generative model — UMAPs of data, trajectories, NB mixture, network modes and simulation; cell-type proportions; gene-pair correlations; proteins
2. Gene regulatory network — regulatory power (violin plots), top-20 regulators + stimulus and top-20 regulated genes (mutual top-10 edges)
3. In-silico perturbations — overview across KO/OV, then one page per perturbation
4. Proliferation — prior vs learned net rates, population growth, proteins driving growth; protein dilution at the birth rate
5. Learned dynamics — mRNA and protein fields (mechanistic velocity, displacement along trajectories), summary on mRNA
6. Held-out test cells — predictions with the network fixed vs the test data

Data — UMAP of all the cells

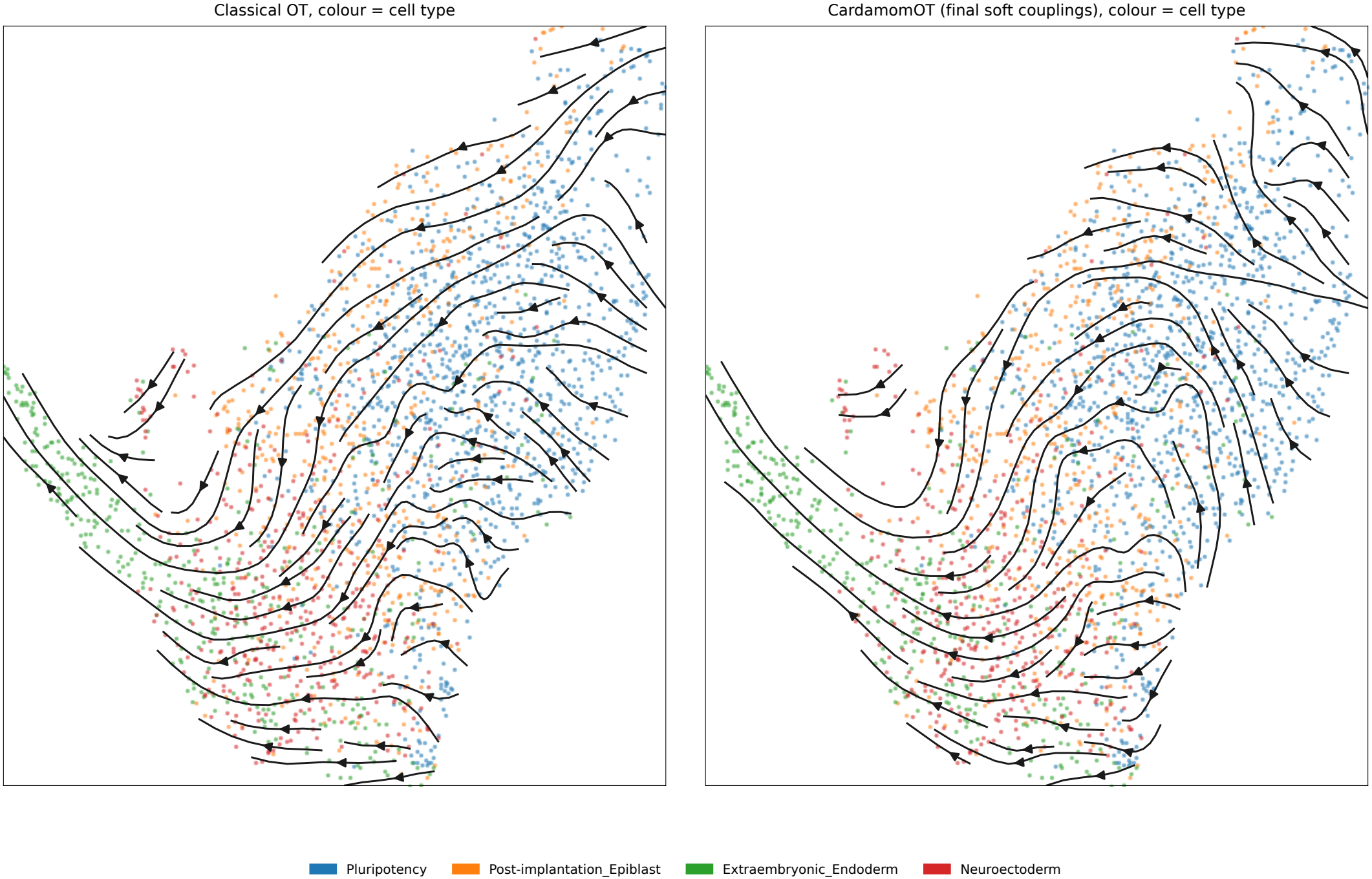
Data/data.h5ad, 2449 cells: log1p(counts), 41 highly variable genes, PCA (40 components), kNN graph, UMAP. Stored in cardamomOT/embedding_data.npz.





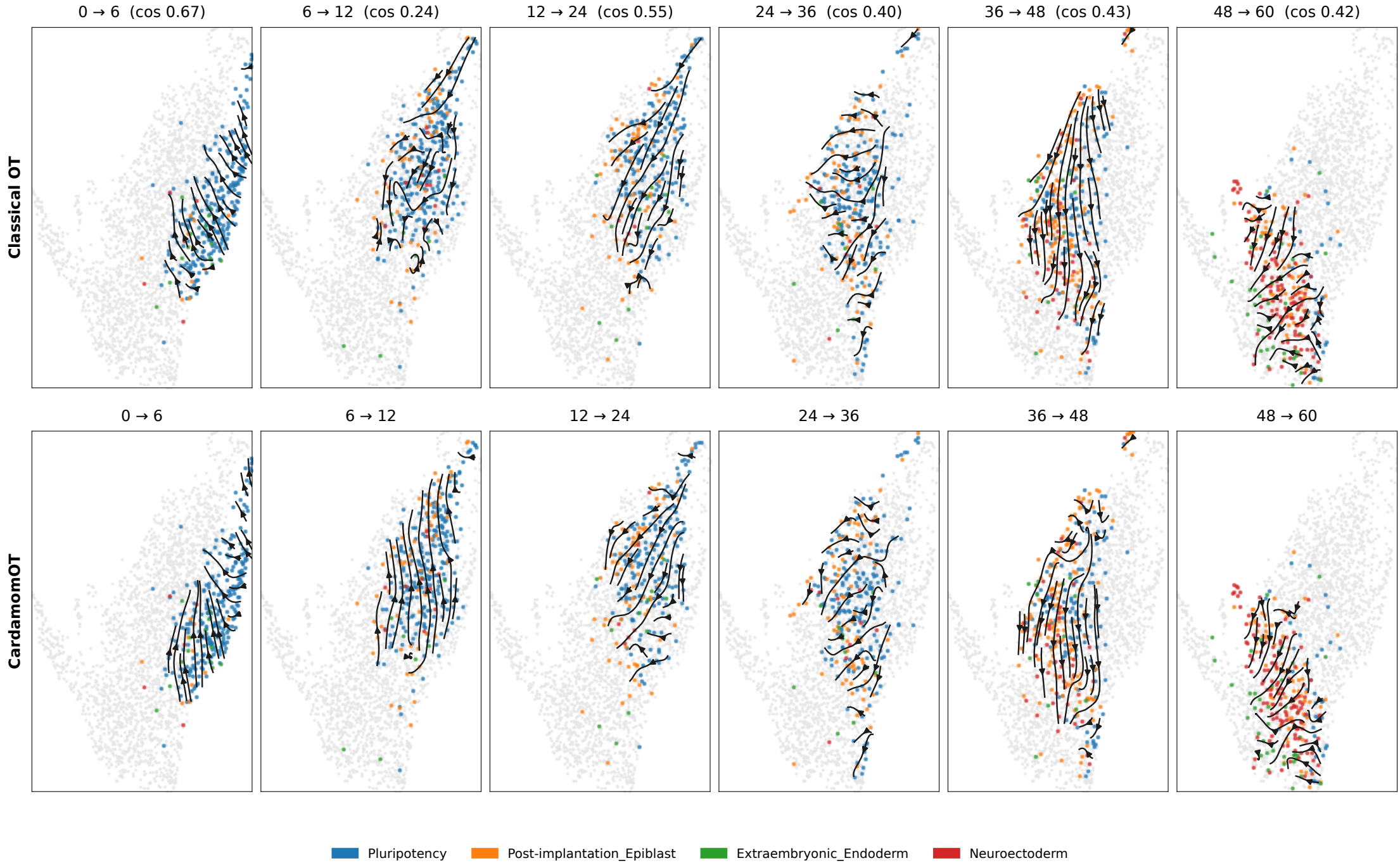
Teaser — displacement fields of classical OT and CardamomOT

Displacement to the mean descendant (x(t+1) − x(t), not divided by Δt), all times. Agreement on the cells with both direct displacements: weighted cosine 0.55 (PCA space). CardamomOT: final soft couplings of the trajectories (1714 cells); cells never reached or outside the training set: Gaussian kernel on the model mRNAs.



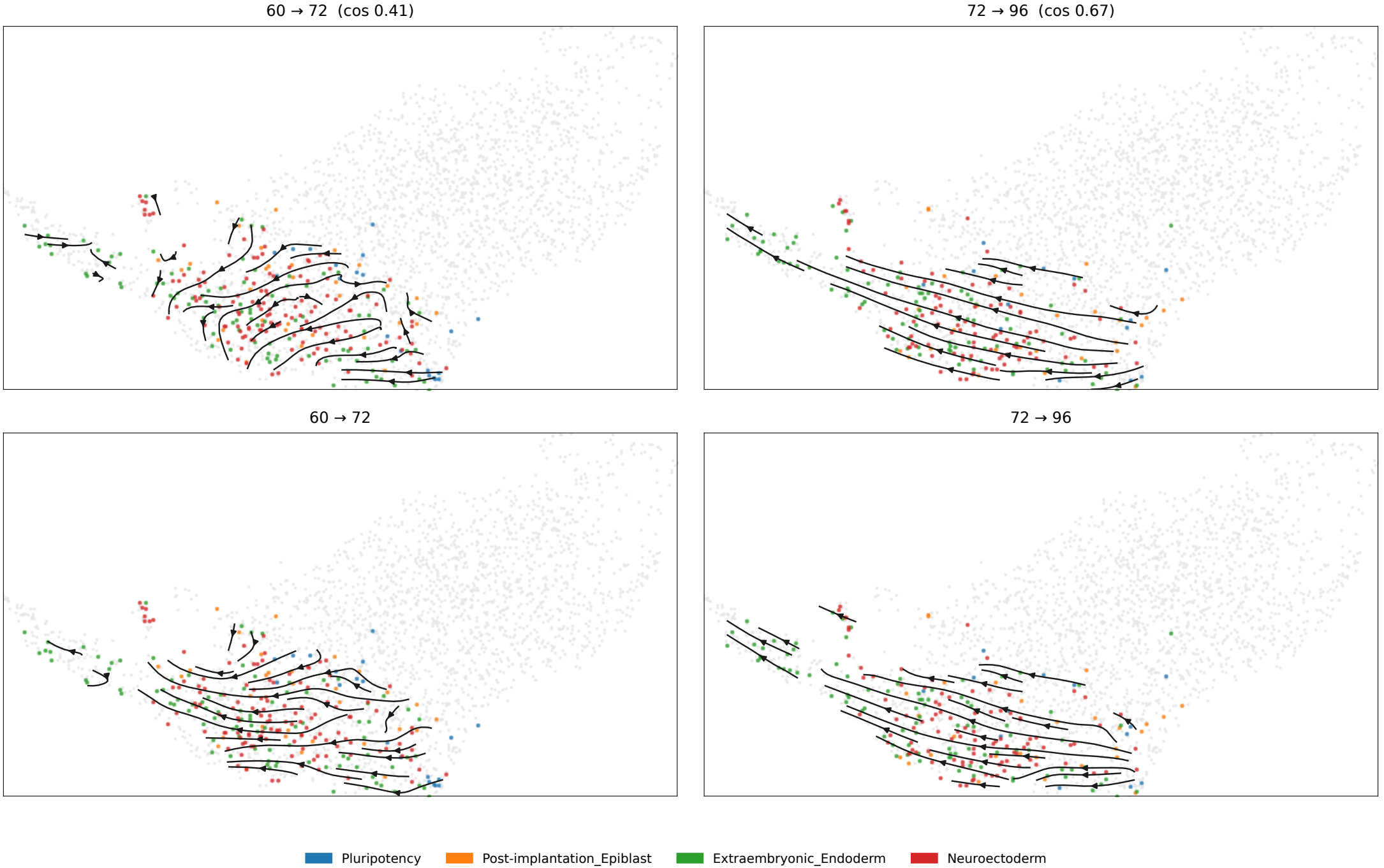
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) towards their mean descendant at t_to, kNN projection among the cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



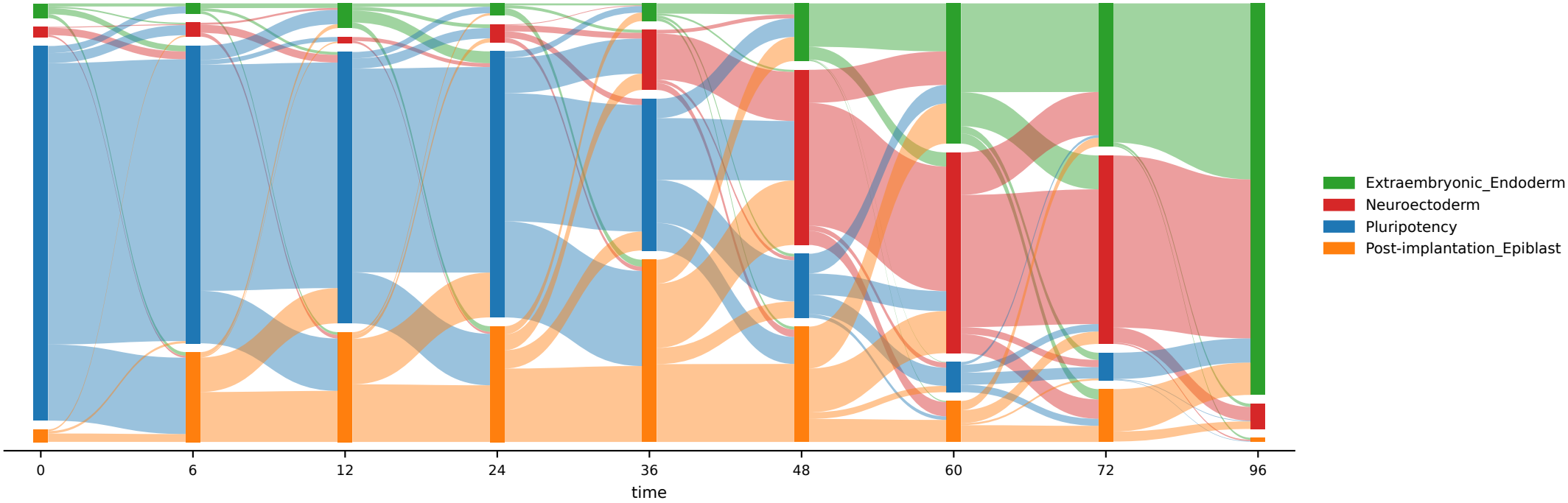
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) towards their mean descendant at t_to, kNN projection among the cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



Classical OT — cell-type transitions and fate genes

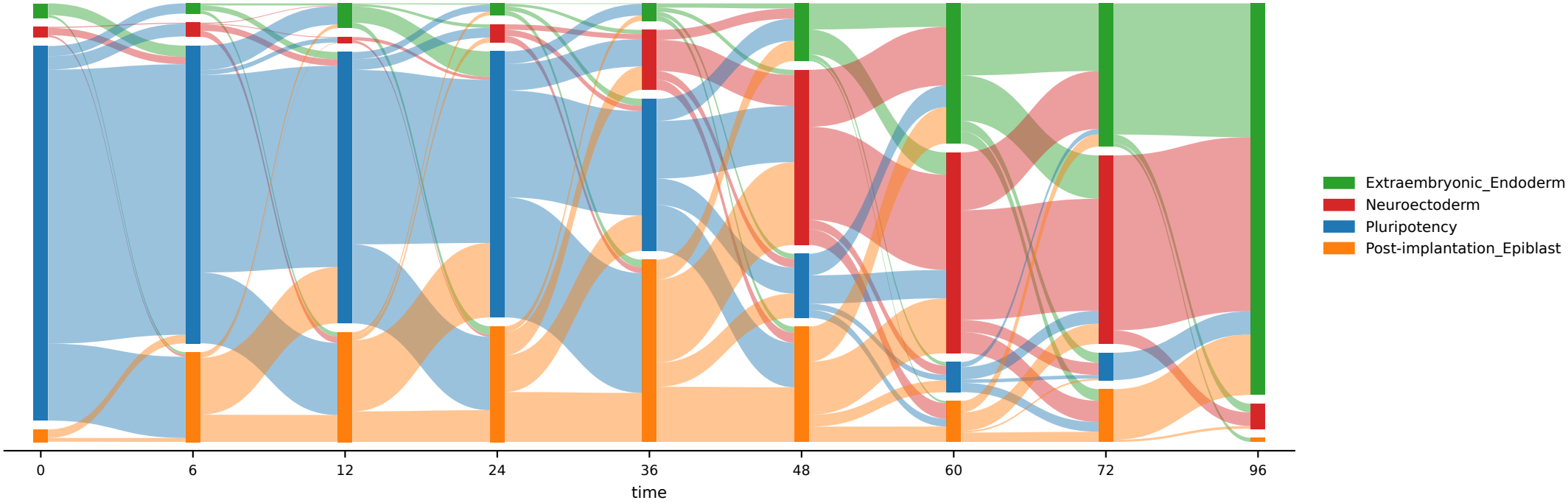
Waddington-OT couplings (every gene, train cells, per sample, growth from the net rate), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (classical_OT/fate_genes.csv).



	Top-10 genes up in the ancestors (vs ancestors of the other cells)										Top-10 genes up in the cell type (vs the other cells)									
Extraembryonic_Endoderm	**Sparc**	**Lamb1**	**Col4a2**	Cdh2	Cd24a	**Dab2**	**Krt8**	**Gata6**	Sox11	Meis1	**Sparc**	**Lamb1**	**Col4a2**	**Dab2**	**Gata6**	**Krt8**	Fst	Pdgfra	Sox17	Gata4
Neuroectoderm	**Hoxb2**	**Hoxa1**	**Sox1**	**Sox11**	**Cd24a**	**Pax6**	**Meis1**	**Cdh2**	Pou3f1	Dnmt3a	**Cd24a**	**Hoxb2**	**Sox11**	**Sox1**	**Cdh2**	**Meis1**	**Hoxa1**	**Pax6**	Nes	Sox9
Pluripotency	**Zfp42**	**Esrrb**	**Pou5f1**	**Klf2**	**Fgf4**	**Klf4**	**Sox2**	**Nr0b1**	**Jarid2**	**Dppa3**	**Pou5f1**	**Zfp42**	**Esrrb**	**Fgf4**	**Klf2**	**Sox2**	**Klf4**	**Jarid2**	**Nr0b1**	**Dppa3**
Post-implantation_Epiblast	**Dppa2**	Zfp42	**Dppa4**	**Dnmt3b**	**Dnmt3a**	**Zic3**	**Zic2**	**Jarid2**	Klf2	Klf4	**Dppa2**	**Dppa4**	**Dnmt3a**	**Dnmt3b**	**Zic2**	**Zic3**	Pou3f1	Wnt8a	**Jarid2**	Cer1

Teaser — CardamomOT: cell-type transitions and fate genes

Final soft couplings of the CardamomOT trajectories (model genes, data_<split> cells), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (cardamomOT/teaser_fate_genes.csv).



	Top-10 genes up in the ancestors (vs ancestors of the other cells)	Top-10 genes up in the cell type (vs the other cells)
Extraembryonic_Endoderm	**Col4a2** Cdh2 **Lamb1** Cd24a **Dab2** Meis1 Sox11 Hoxb2 **Krt8** **Gata6**	Sparc **Lamb1** **Col4a2** **Dab2** **Gata6** **Krt8** Pdgfra Fst Sox17 Gata4
Neuroectoderm	**Hoxb2** **Hoxa1** **Sox1** **Sox11** Dnmt3a **Pax6** Pou3f1 Lamb1 **Meis1** Gata6	**Hoxb2** Cd24a **Sox1** **Sox11** Cdh2 **Meis1** **Hoxa1** **Pax6** Nes Sox9
Pluripotency	**Zfp42** **Klf2** **Esrrb** **Pou5f1** **Fgf4** **Klf4** **Sox2** **Nr0b1** **Jarid2** **Dppa3**	**Pou5f1** **Zfp42** **Esrrb** **Klf2** **Fgf4** **Sox2** **Klf4** **Nr0b1** **Jarid2** **Dppa3**
Post-implantation_Epiblast	Zfp42 Pou5f1 Esrrb **Dppa2** **Dppa4** **Jarid2** **Zic3** **Dnmt3a** **Dnmt3b** Klf4	**Dppa2** **Dppa4** **Dnmt3a** **Dnmt3b** Zic2 Pou3f1 **Zic3** Wnt8a **Jarid2** Fgf5

Genes of the model (41)

Data/data_train.h5ad, alphabetical order.

Ascl1
Cd24a
Cdh2
Cer1
Col4a2
Dab2
Dnmt3a
Dnmt3b
Dppa2
Dppa3
Dppa4
Esrrb
Fgf4
Fgf5
Fst
Gata4
Gata6
Hoxa1
Hoxb2
Jarid2
Klf2
Klf4
Krt8
Lamb1
Meis1
Nes
Nr0b1
Pax6
Pdgfra
Pou3f1
Pou5f1
Sox1
Sox11
Sox17
Sox2
Sox9
Sparc
Wnt8a
Zfp42
Zic2
Zic3

0. Per-cell sequencing depth

Diagnostic on the whole transcriptome within homogeneous groups (sample, time, cell type); counts are modelled as NB(k, c / s_i) when the depth factor s_i is applied.

Depth

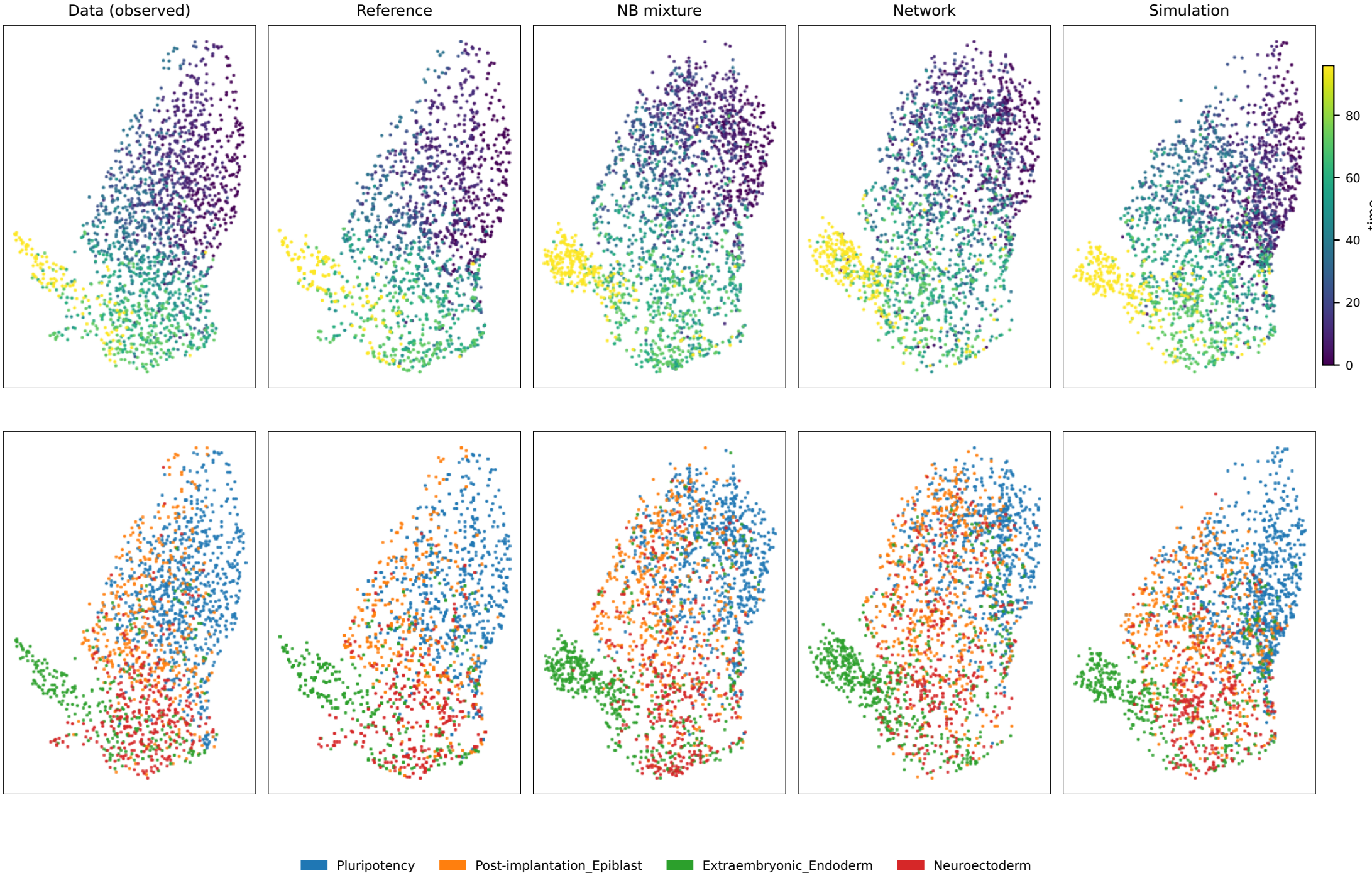
not estimated

Reason

41 genes < 10000

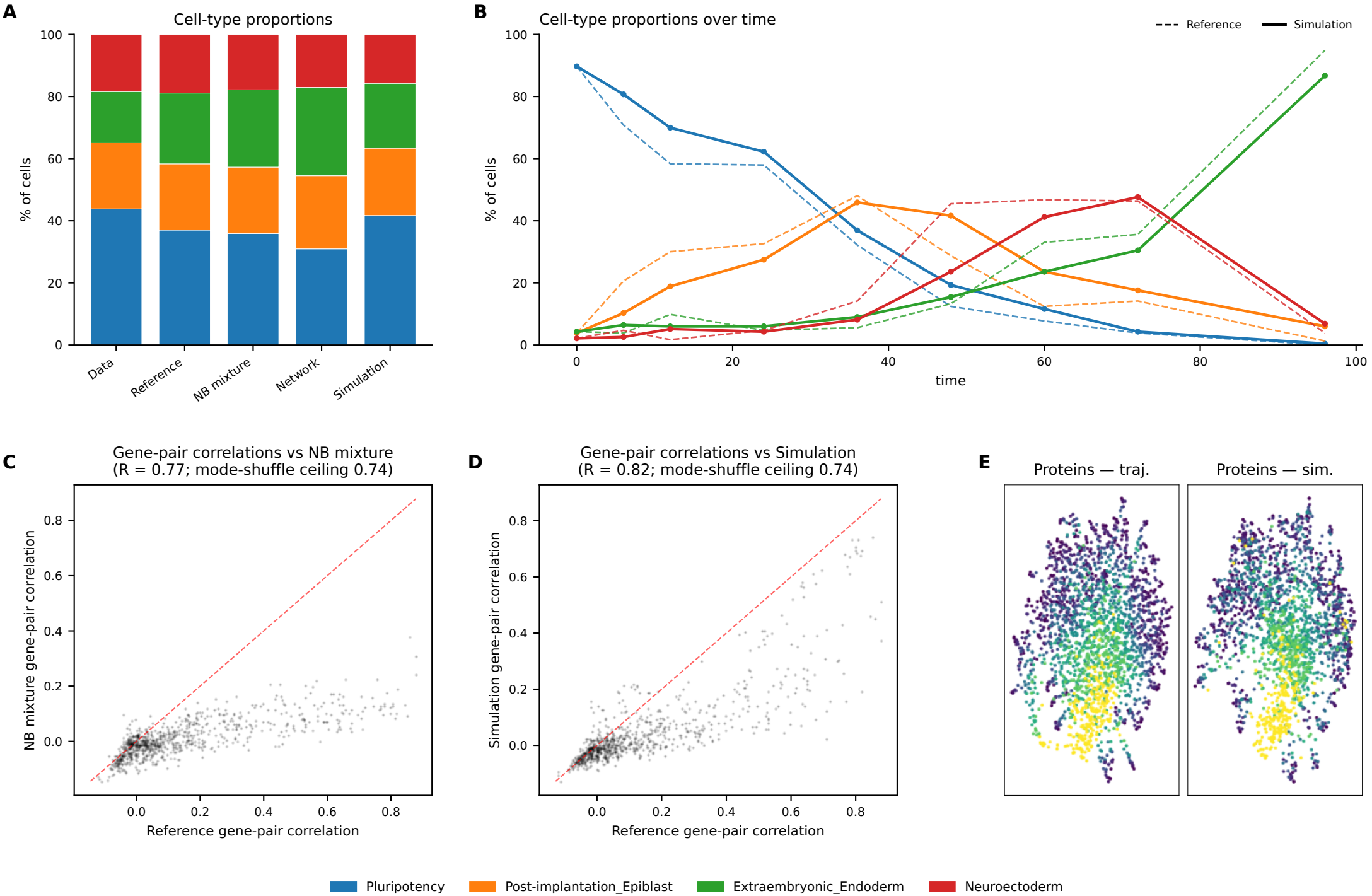
1. Generative model — trajectories and simulation

UMAP learned on the observed data; the trajectories (Reference: one state per ancestor, i.e. without proliferation, as the simulation), the NB mixture and network-driven modes along them and the full simulation are projected onto it.



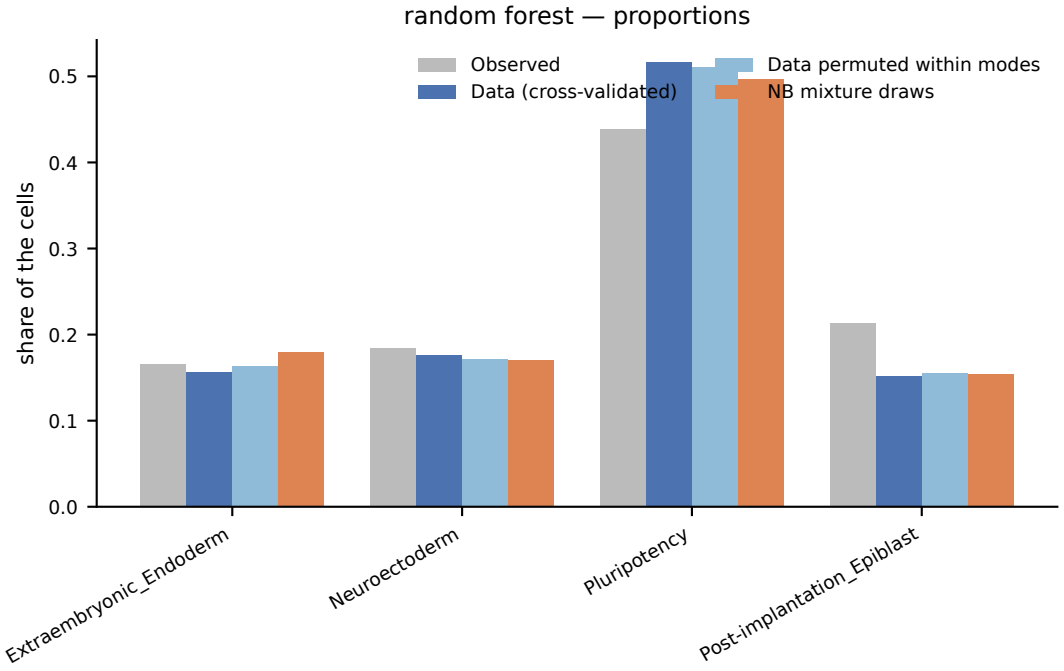
1. Generative model — quantitative checks

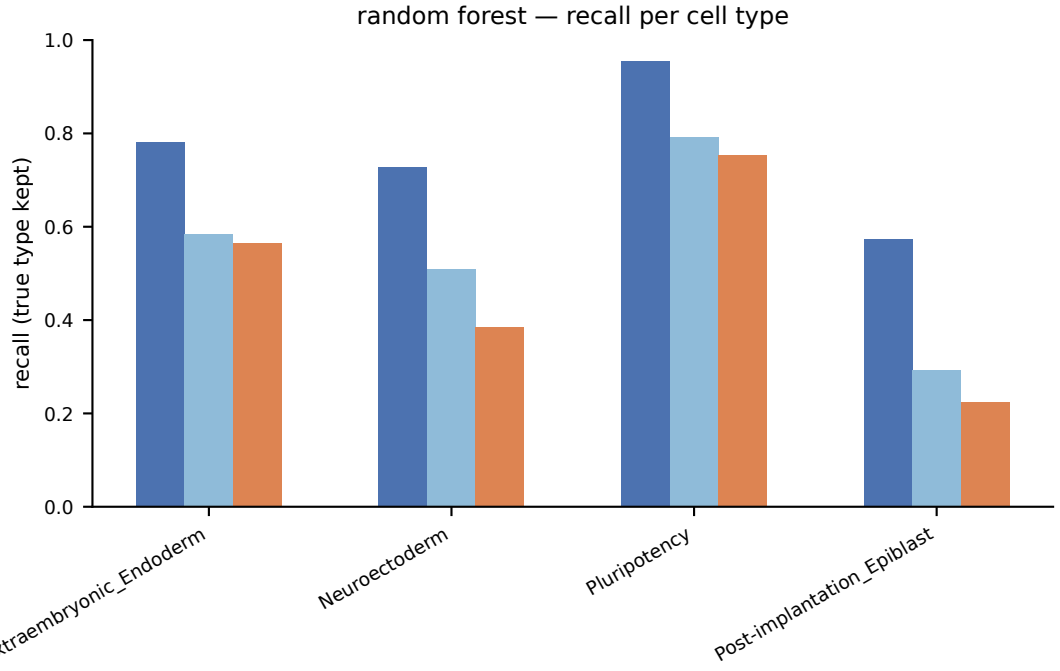
Reference = RNA of the trajectories equivalent to the simulation (without proliferation).

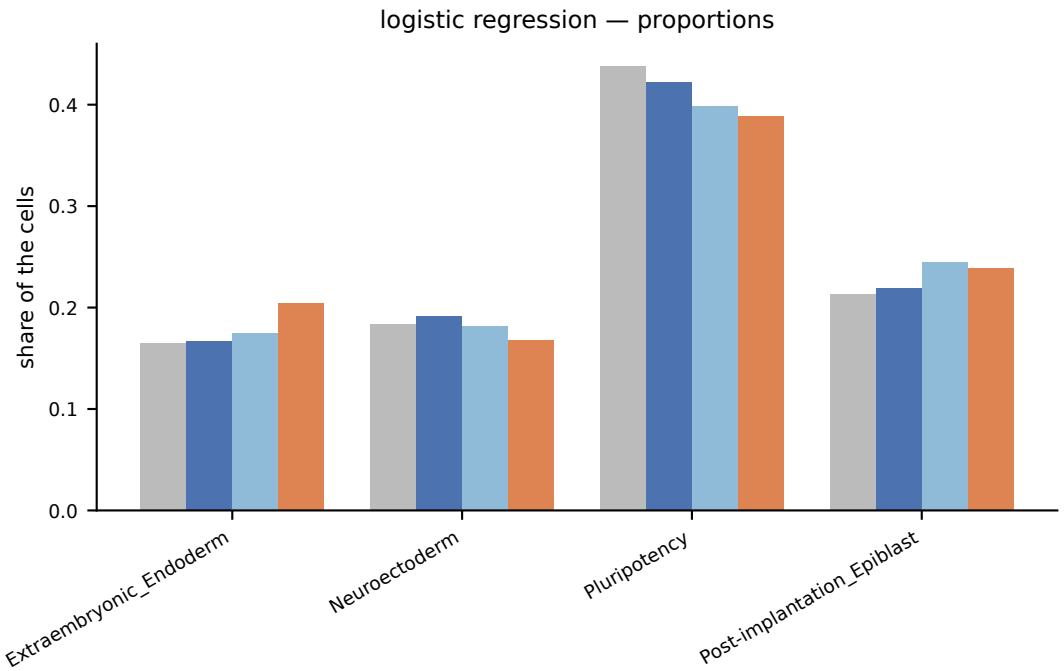


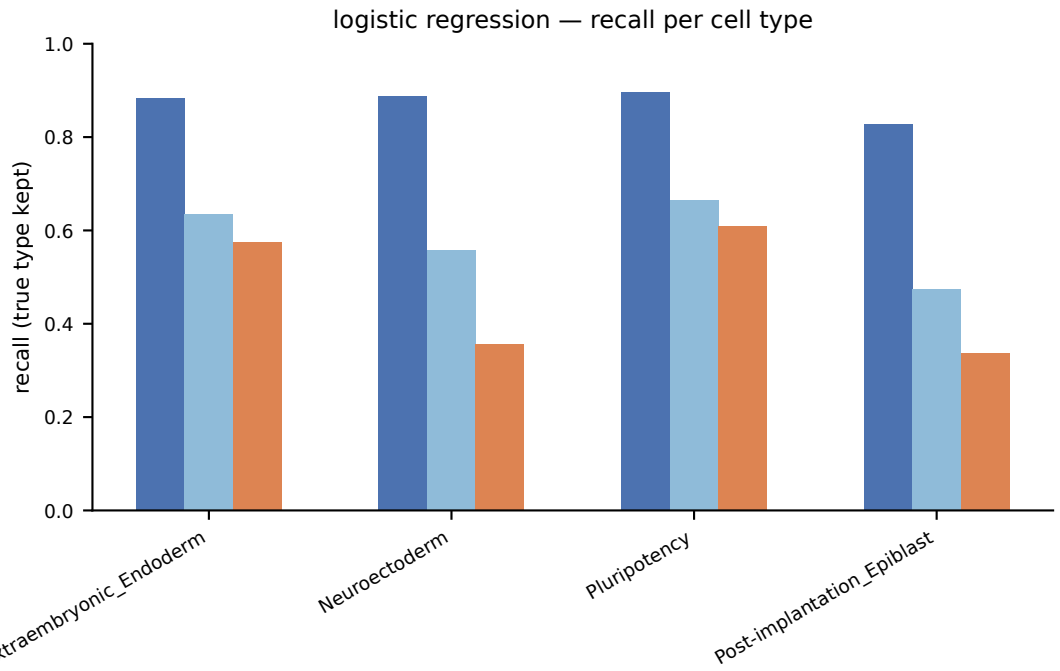
1. Generative model — cell-type identity in the NB mixture

One draw per real cell, per-sample classifiers trained on the other cells. Cross-validated data = classifier limit; permuted within modes (same sample, time, mode) = best model with these modes. Gap data → permuted: modes too coarse; permuted → draws: NB fit.







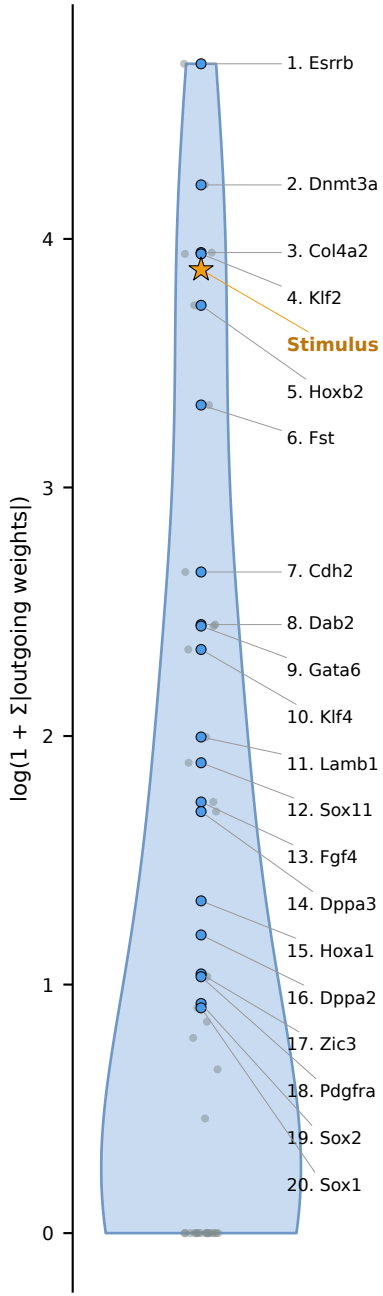


2. Gene regulatory network — regulatory power

Self-interactions excluded. Top 20 genes labelled; stimulus shown as a star (stimulus = 1.0 ≥ 0.5).

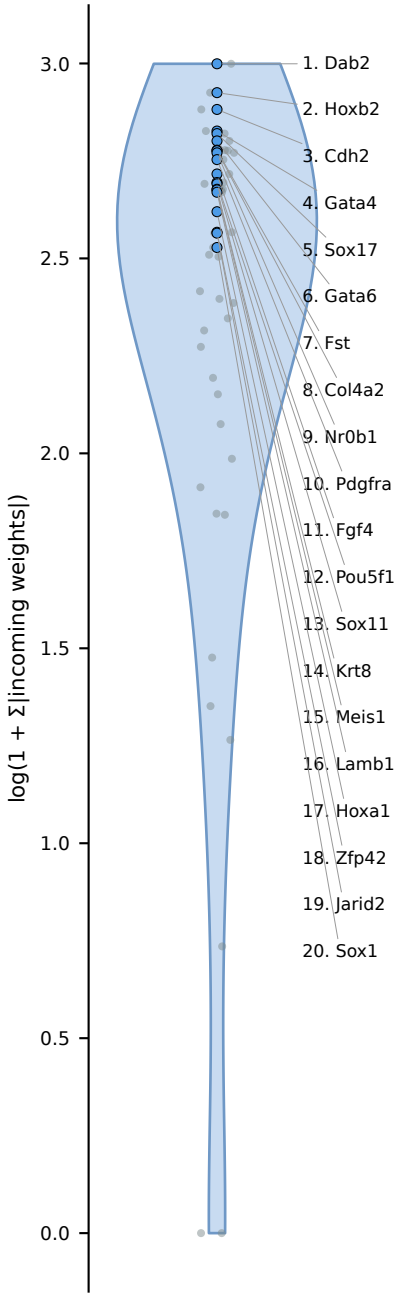
A

Outgoing regulation (regulators)



B

Incoming regulation (targets)



C

Top 20 regulators

#	Regulator	Σ|w|	act.	inh.
1	Esrrb	109.46	14	15
2	Dnmt3a	66.85	12	11
3	Col4a2	50.68	10	1
4	Klf2	50.40	8	15
5	Hoxb2	40.80	17	13
6	Fst	26.99	6	6
7	Cdh2	13.29	12	0
8	Dab2	10.57	11	1
9	Gata6	10.50	14	5
10	Klf4	9.47	3	10
11	Lamb1	6.36	5	1
12	Sox11	5.63	12	0
13	Fgf4	4.67	1	8
14	Dppa3	4.45	4	6
15	Hoxa1	2.81	5	1
16	Dppa2	2.32	3	1
17	Zic3	1.84	1	2
18	Pdgfra	1.81	4	0
19	Sox2	1.52	3	1
20	Sox1	1.48	6	0
★	Stimulus	47.23	13	8

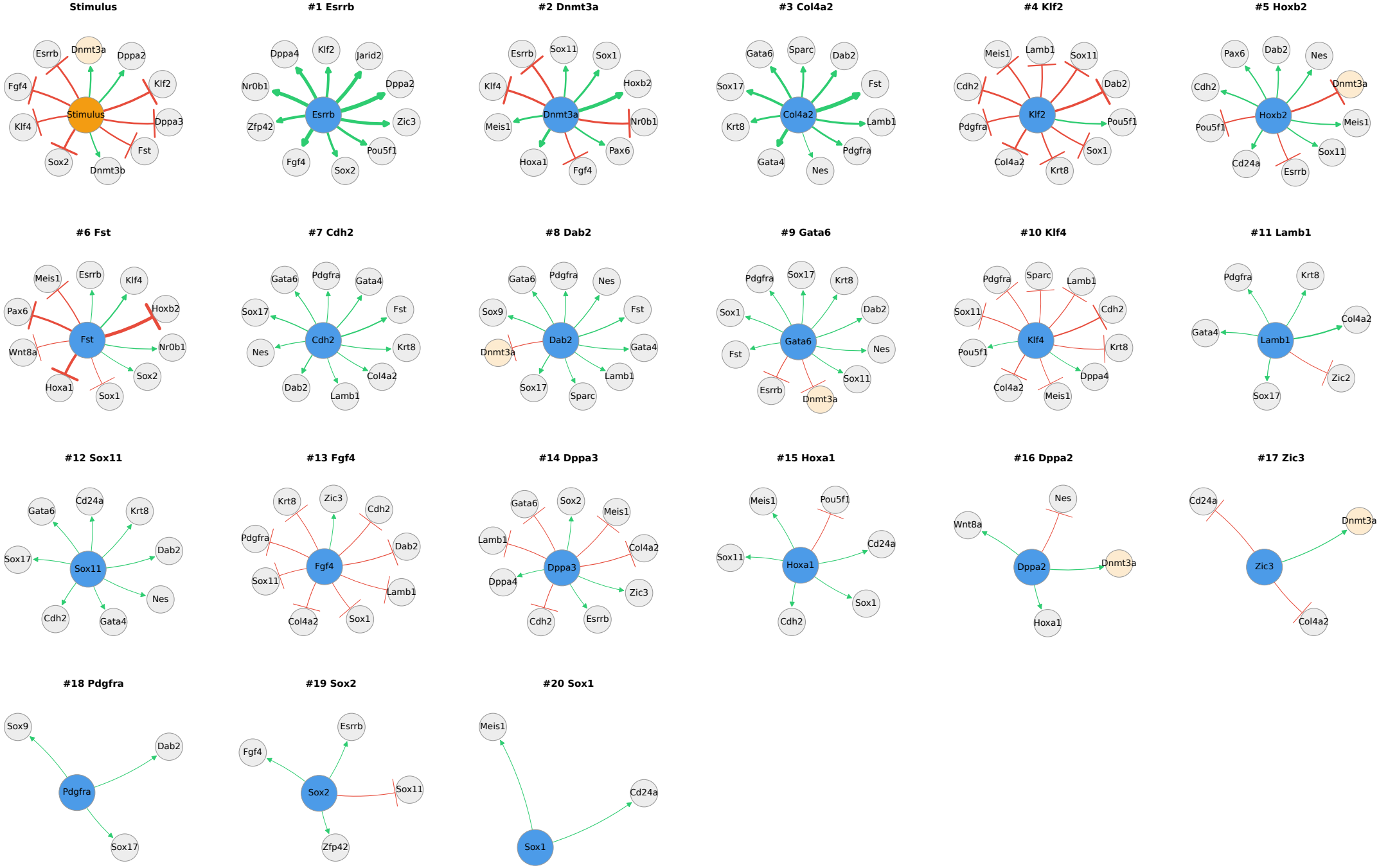
D

Top 20 regulated genes

#	Target	Σ|w|	act.	inh.
1	Dab2	19.07	8	3
2	Hoxb2	17.64	1	2
3	Cdh2	16.85	7	5
4	Gata4	15.89	7	2
5	Sox17	15.78	10	2
6	Gata6	15.47	8	4
7	Fst	15.08	5	0
8	Col4a2	15.06	7	6
9	Nr0b1	14.97	2	2
10	Pdgfra	14.70	9	5
11	Fgf4	14.13	4	3
12	Pou5f1	13.80	3	4
13	Sox11	13.75	5	5
14	Krt8	13.55	7	4
15	Meis1	13.52	5	5
16	Lamb1	13.44	6	5
17	Hoxa1	12.73	2	1
18	Zfp42	12.03	4	2
19	Jarid2	12.00	2	2
20	Sox1	11.53	4	5

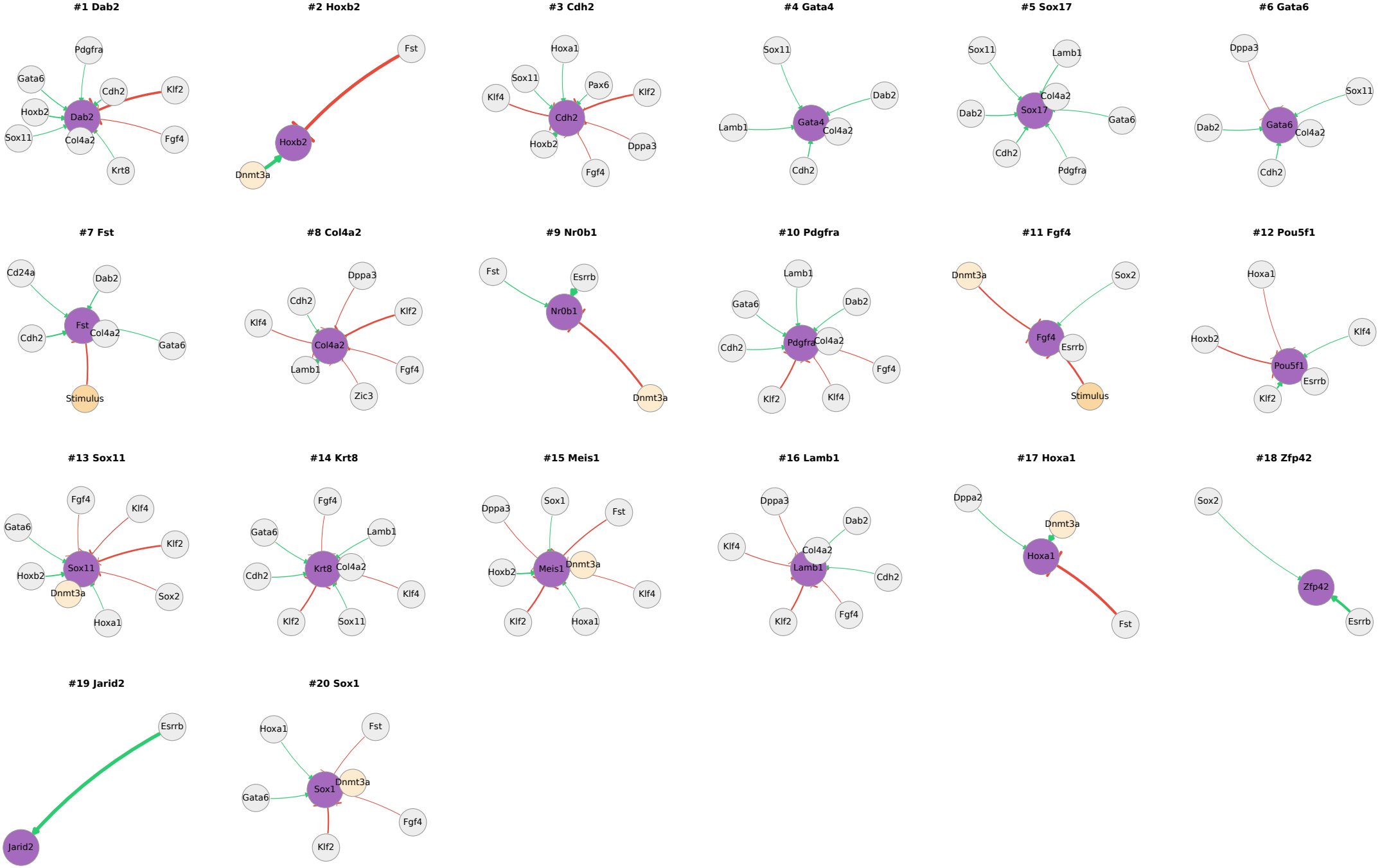
2. Gene regulatory network — top regulators and their main targets

Edges shown only if among the 10 strongest (|w|) targets of the regulator AND the 10 strongest regulators of the target; edge width ∝ |weight| (global scale).
Perturbed genes (KO/OV) are highlighted in beige.



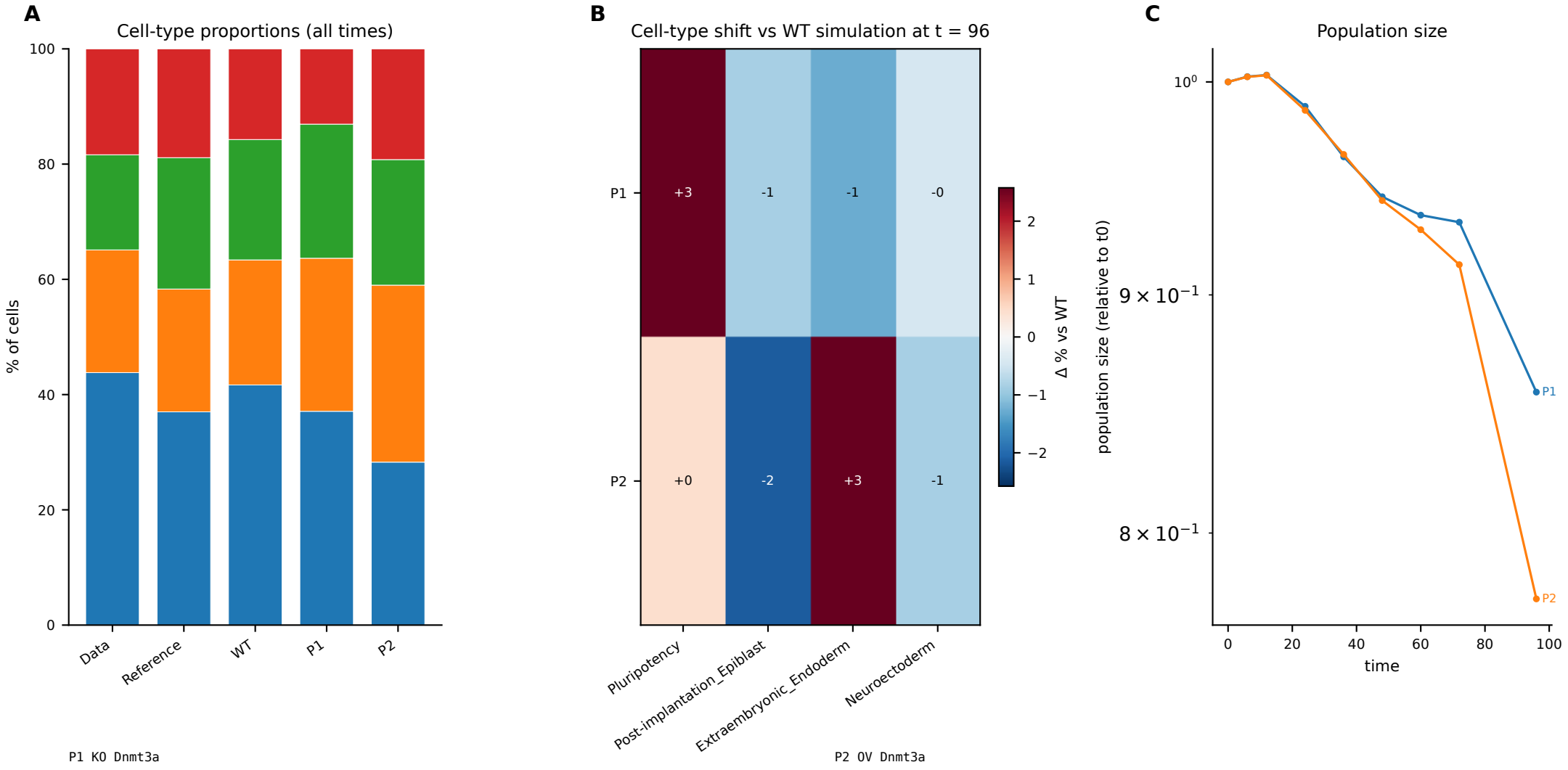
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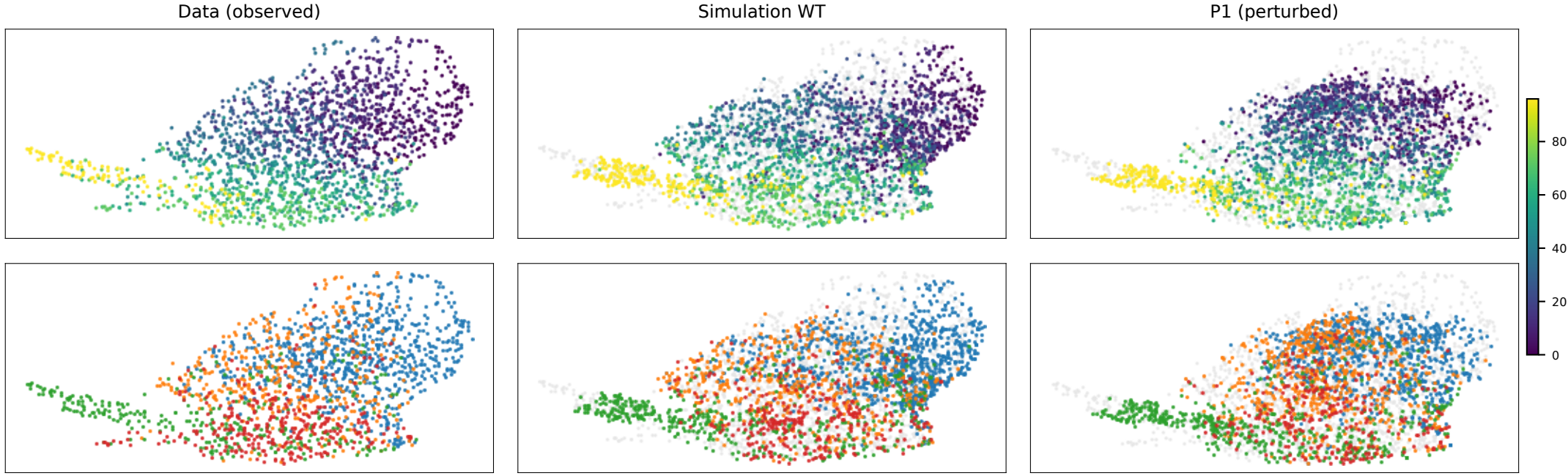
3. In-silico perturbations — overview

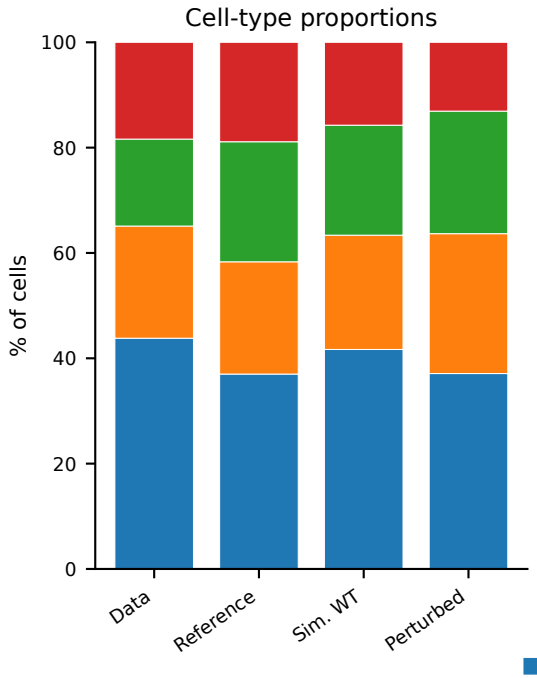
Cell types predicted by per-sample classifiers (logistic) trained on the observed data. Population sizes from the branching simulation (proliferation MLP + RATE effects).

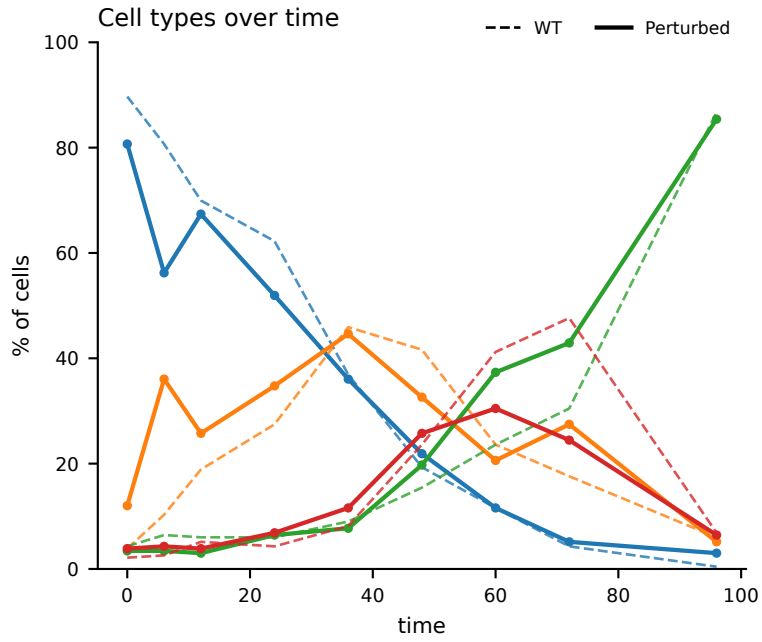


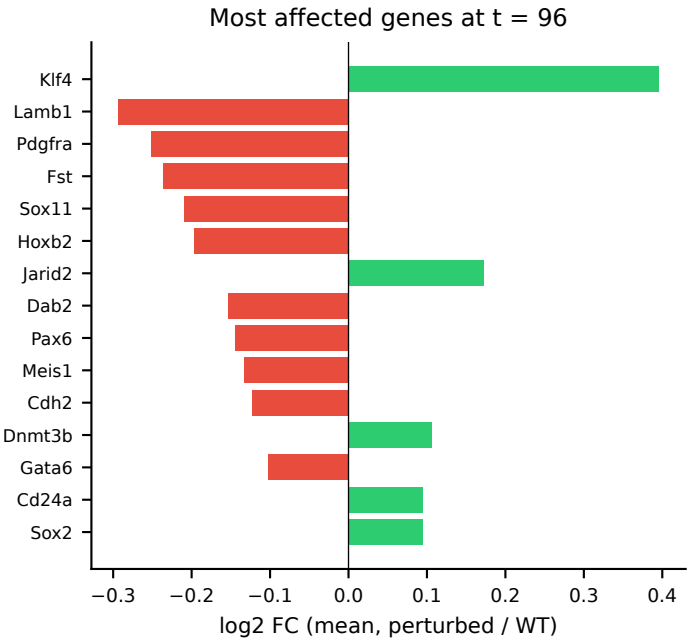
3. Perturbation P1 — KO Dnmt3a

KO_Dnmt3a_OV_none. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).





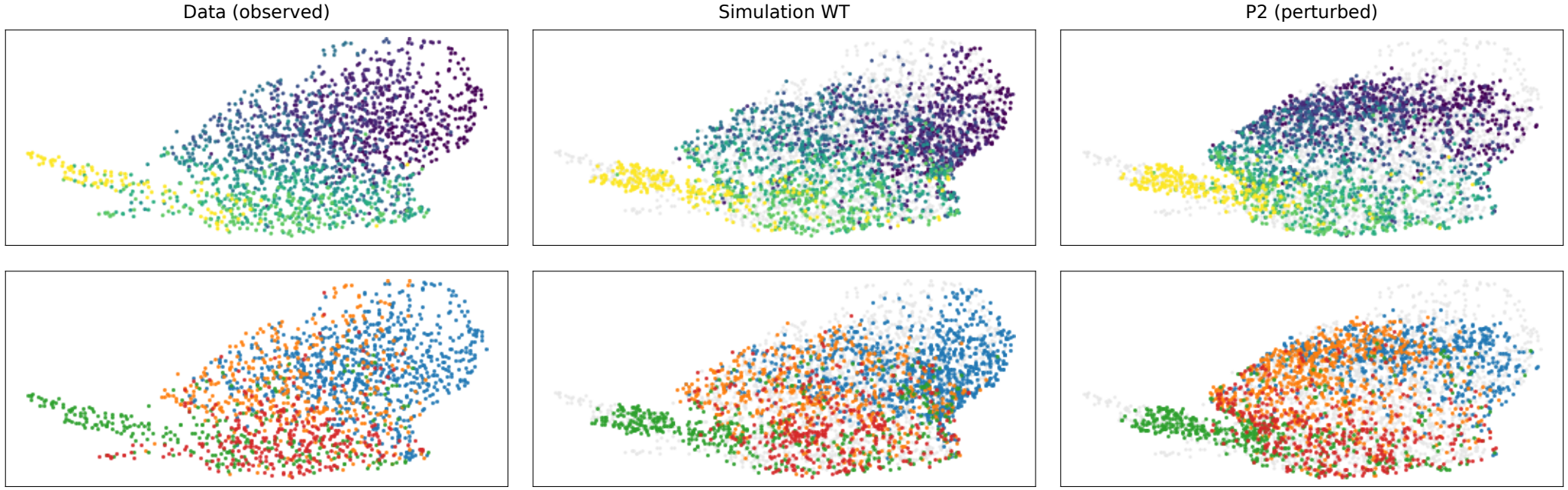


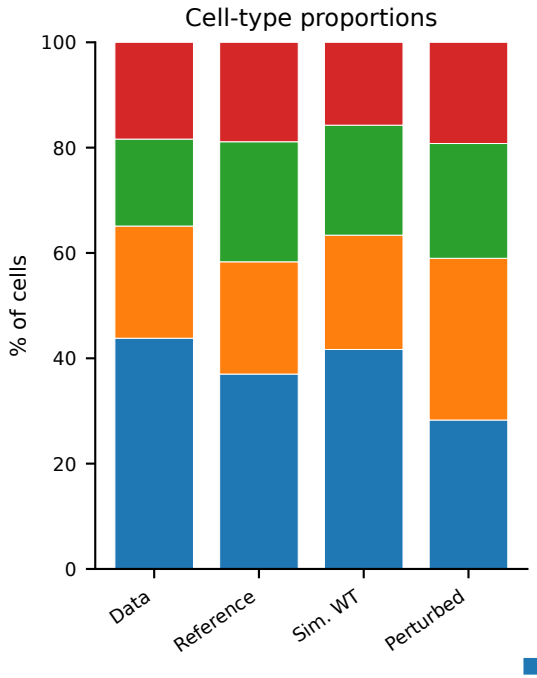


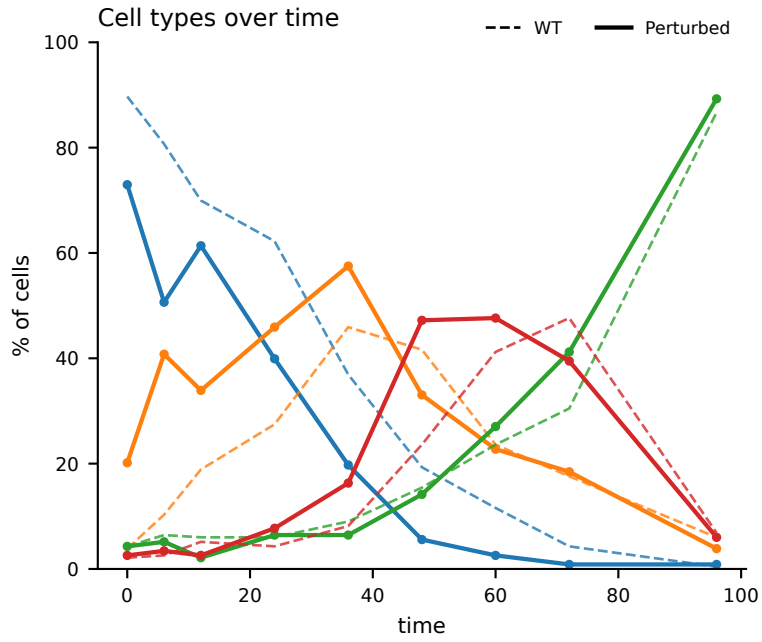
Pluripotency Post-implantation_Epiblast Extraembryonic_Endoderm Neuroectoderm

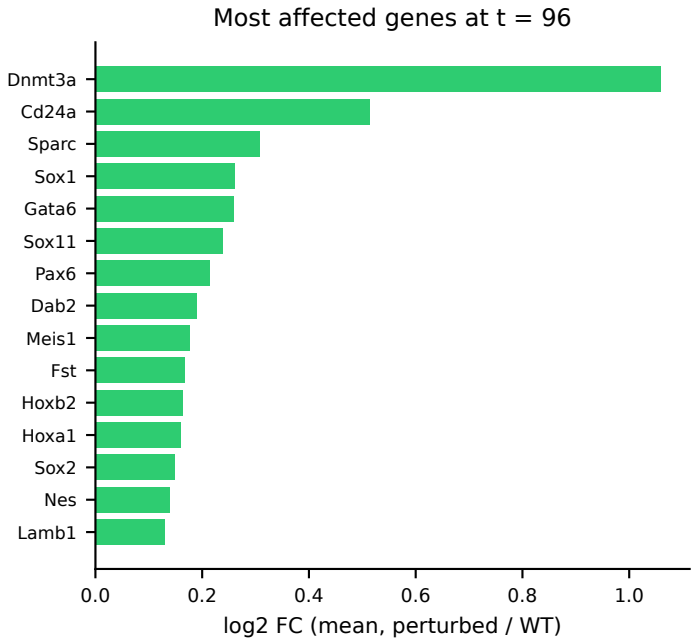
3. Perturbation P2 — OV Dnmt3a

KO_none_OV_Dnmt3a. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).











4. Proliferation

Section unavailable:

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FileNotFoundError('no growth estimate (data_R_opt.npy / data_R_prior.npy from infer_network_structure)')
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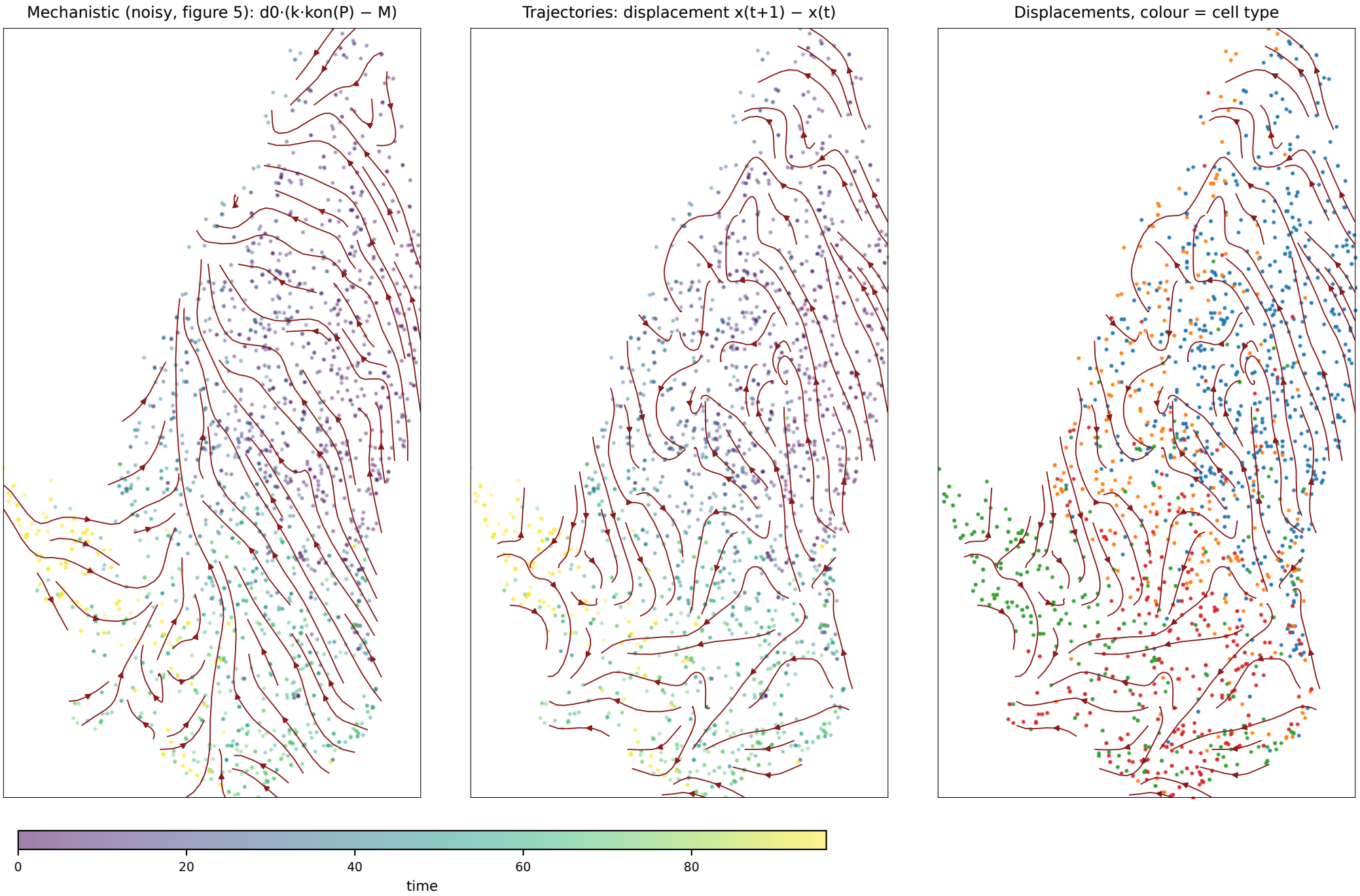
4. Protein dilution

Section unavailable:

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ValueError('protein_dilution = False, or no birth rate (obs proliferation_birth_rate / net rate)')
```

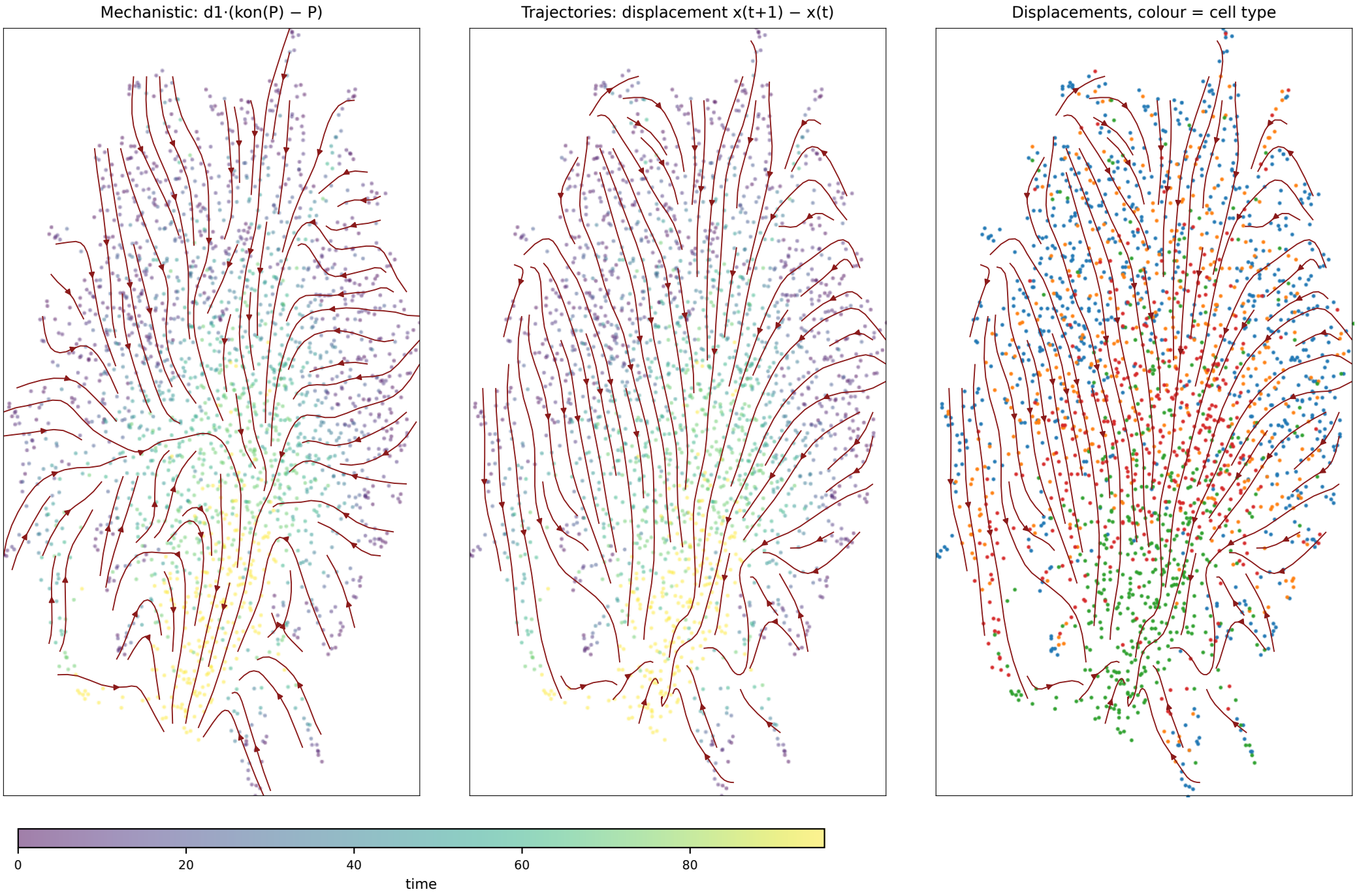
5. Learned dynamics — mRNA velocity and displacement fields

All 2097 trajectory states projected onto the UMAP learned on the observed mRNA (log1p, raw counts); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.08.



5. Learned dynamics — Proteins velocity and displacement fields

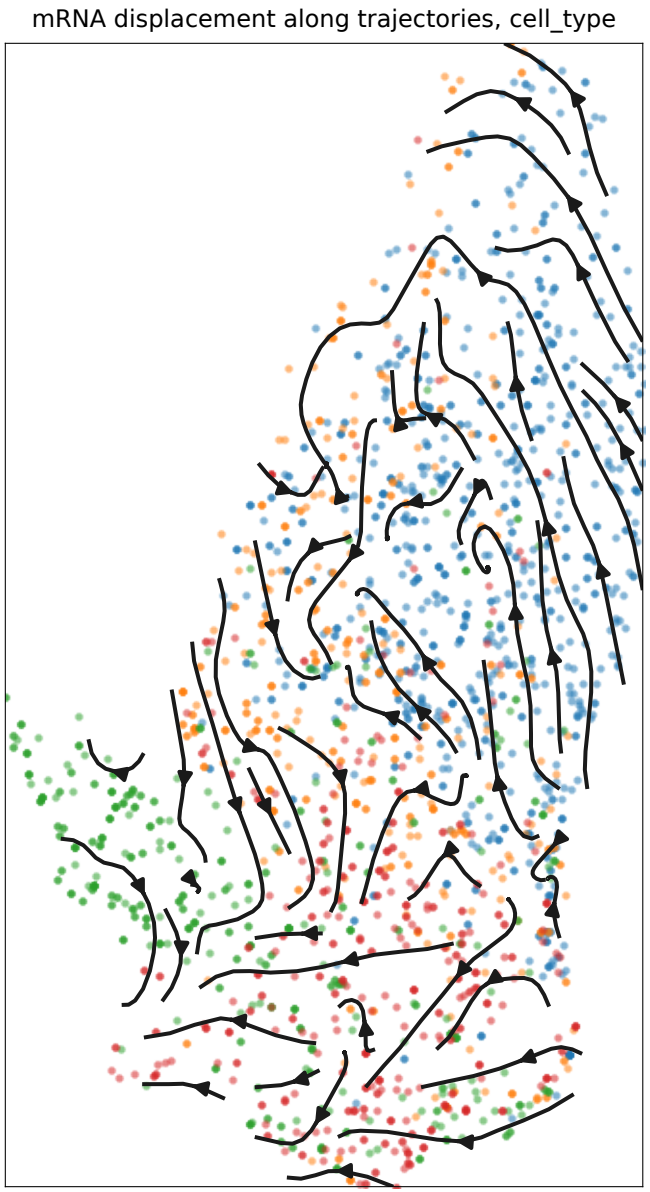
All 2097 trajectory states projected onto the UMAP learned on the protein trajectories (raw levels); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.27.



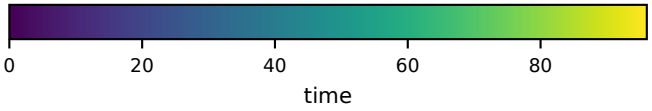
5. Learned dynamics — summary

All 2097 trajectory states on the UMAP learned on the observed mRNA (log1p, raw counts); displacement along the inferred trajectories x(t+1) − x(t); background of the field: cell_type.





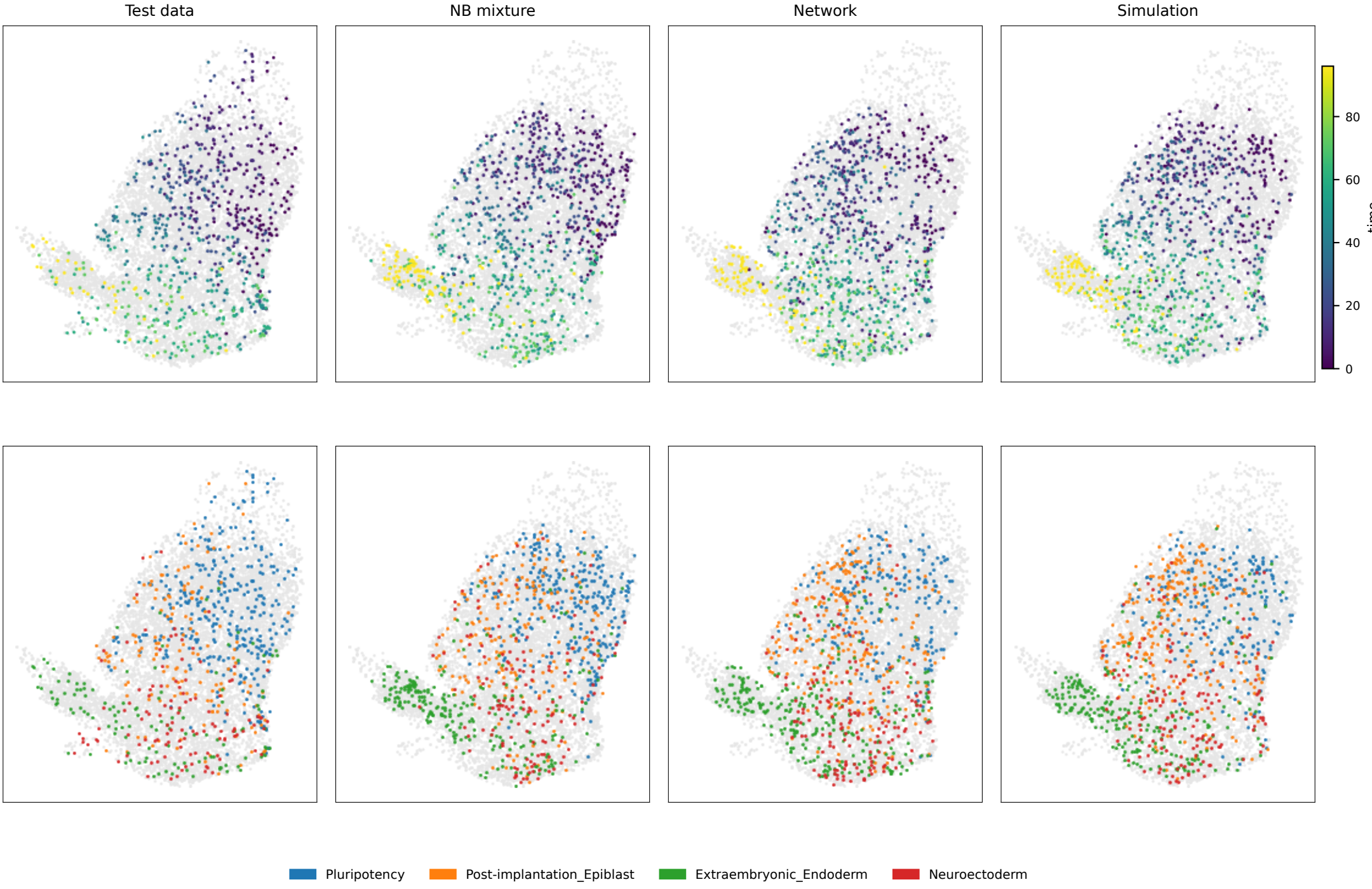
No net proliferation rate





6. Held-out test cells — predictions with the network fixed

Test cells classified into basins with the training mixtures, trajectories inferred with the training network fixed, then simulated; projected onto the UMAP learned on the observed data (grey: training).



6. Held-out test cells — quantitative checks

Per gene: 1-D Wasserstein distance between log1p counts, averaged over the common timepoints.

